Data Sharing Statement: See Supplement 2.

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Clinical Characteristics of Erythema Nodosum and Associations With Chronicity and Recurrence

Erythema nodosum (EN), the most common panniculitis, classically presents as erythematous nodules in the bilateral pretibial areas. Literature characterizing EN is scarce—restricted to small (N < 200) cohorts outside of the US, with fewer data examining chronicity/recurrence.[1-5] In this retrospective cohort study, we characterize the presentation, causes, treatment, and disease course of EN, and identify associations with chronicity/recurrence.

Methods | After obtaining institutional review board exemption, we queried medical records at New York University Langone Health for EN, manually reviewing records for diagnostic accuracy. Patients aged 18 to 89 years with follow-up longer than 12 weeks presenting with EN from January 1, 2000, to November 30, 2021, were included. Patients with diagnoses of other panniculitides (including panniculitis not otherwise specified) were excluded. Data were extracted on demographics, presentation, workup, triggers, comorbidities, treatments, and response. Patients were classified as having classic EN if symptoms resolved within 12 weeks and chronic/recurrent EN if symptoms persisted/recurred 12 weeks or longer following initial onset. We analyzed variables with Pearson χ^2 test, Fisher exact test, unpaired *t* test, or Wilcoxon rank sum test using R, version 4.2.2 (R Foundation for Statis-

Supplemental content

Table 1. Demographics, Clinical Features, and Treatment of Patients With Classic Erythema Nodosum (EN) and Chronic/Recurrent EN

Characteristic	No. (%) All patients	Classic EN	Chronic/recurrent EN	*P* value
Total	340	181	159	NA
Age at onset, median (IQR), y	41 (29-56)	42 (21-58)	39 (27-51)	.02
Sex				
Female	297 (87)	153 (85)	144 (91)	.09
Male	43 (13)	28 (15)	15 (9)	
Race and ethnicity[a]				
Asian	17 (5)	9 (5)	8 (5)	.24
Black or African American	56 (16)	25 (14)	31 (19)	
White	205 (60)	118 (65)	87 (55)	
Other or unknown	62 (18)	29 (16)	33 (21)	
Lesions on bilateral lower extremities only	206 (61)	109 (60)	97 (61)	.88
Lesions involving nonleg locations				
Head and neck	7 (2)	5 (3)	2 (1)	.46
Upper extremities	58 (17)	29 (16)	29 (18)	.59
Trunk	17 (5)	5 (3)	12 (8)	.04
Extracutaneous symptoms[b]	179 (53)	96 (53)	83 (52)	.88
Workup completed				
Assessed by dermatologist or rheumatologist	251 (74)	119 (66)	132 (83)	<.001
Autoantibodies[c]	213 (63)	100 (55)	113 (71)	.003
Chest radiography	165 (49)	79 (44)	86 (54)	.05
Colonoscopy/endoscopy	32 (9)	9 (5)	23 (14)	.003
Human leukocyte antigen B27	38 (11)	15 (8)	23 (14)	.07
Hepatitis B/C or HIV	126 (37)	53 (29)	73 (46)	.002
Lyme disease	77 (23)	28 (21)	39 (25)	.44
Tuberculosis	148 (44)	62 (34)	86 (54)	<.001
Tissue biopsy	106 (31)	33 (18)	73 (46)	<.001
Tissue culture	8 (2)	2 (1)	6 (4)	.15

(continued)

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Table 1. Demographics, Clinical Features, and Treatment of Patients With Classic Erythema Nodosum (EN) and Chronic/Recurrent EN (continued)

Characteristic	No. (%)			P value
	All patients	Classic EN	Chronic/ recurrent EN	
Treatments administered				
No treatment	82 (24)	52 (29)	30 (19)	.03
Acetaminophen	7 (2)	4 (2)	3 (2)	.99
Antibiotics, systemic	31 (9)	21 (12)	10 (6)	.09
Azathioprine	3 (1)	0	3 (2)	.10
Colchicine	33 (10)	4 (2)	29 (18)	<.001
Intralesional corticosteroids	9 (3)	1 (1)	8 (5)	.01
Systemic corticosteroids	130 (38)	64 (35)	66 (42)	.26
Topical corticosteroids	50 (15)	18 (10)	32 (20)	.008
Dapsone	5 (1)	0	5 (3)	.02
Hydroxychloroquine	25 (7)	3 (2)	22 (14)	<.001
Methotrexate	18 (5)	3 (2)	15 (9)	.001
Mycophenolate mofetil/mycophenolic acid	3 (1)	1 (1)	2 (1)	.60
NSAIDs, systemic	125 (37)	64 (35)	61 (38)	.57
NSAIDs, topical	3 (1)	0	3 (2)	.10
Potassium iodide	14 (4)	1 (1)	13 (8)	<.001
Tetracyclines	17 (5)	8 (4)	9 (6)	.60
TNF inhibitors	13 (4)	4 (2)	9 (6)	.10
Other immunomodulating therapy[d]	5 (1)	2 (1)	3 (2)	.67
Other nonimmunomodulating therapy[e]	6 (2)	2 (1)	4 (3)	.42
Any systemic therapy (excluding acetaminophen and NSAIDs)	187 (55)	86 (48)	101 (64)	.003
≥2 Therapies required to achieve clinical response	112 (33)	45 (25)	67 (42)	<.001
Incomplete response to therapy	43 (13)	2 (1)	41 (26)	<.001

Abbreviations: NA, not applicable; NSAIDs, nonsteroidal anti-inflammatory drugs; TNF, tumor necrosis factor.

[a] Race and ethnicity were collected from medical records to explore potential racial or ethnic differences in the disease course of EN and causes or disease associations of EN. Patients were included in the other or unknown category if their race was listed as other in the medical record or no race or ethnicity was listed.

[b] Includes arthralgia, cough, diarrhea, fever, and/or sore throat.

[c] Includes antinuclear antibodies, antineutrophil cytoplasmic antibodies, antiphospholipid antibodies, cyclic citrullinated peptides, centromere antibodies, double-stranded DNA antibodies, histone antibodies, myeloperoxidase antibodies, proteinase 3 antibodies, rheumatoid factor autoantibodies, ribonucleoprotein, SSA antibodies, SSB antibodies, and Smith antibodies. When autoantibodies were analyzed individually, each autoantibody was considerably more likely to have been collected in cases of chronic/recurrent EN.

[d] Includes apremilast, leflunomide, mesalamine, sulfasalazine, ustekinumab, vedolizumab, and 6-mercaptopurine.

[e] Includes vitamin B_{12} injections, pentoxifylline, R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisolone), tonsillectomy, and topical tacrolimus.

tical Computing). $P < .05$ indicated statistical significance. This report followed STROBE reporting guidelines.

Results | Of 740 records, 340 patients with EN confirmed clinically or histopathologically were identified. Classic EN (mean [SD] follow-up, 3.6 [2.6] years) occurred in 181 patients (53%) and chronic/recurrent EN (mean [SD] follow-up, 4.4 [2.7] years) in 159 (47%). Among 120 patients with chronic/recurrent EN with multiple episodes, the median episode number was 3, and the median time between first and last EN episode was 3.4 years.

Compared with classic EN, patients with chronic/recurrent EN were more likely to have younger age at symptom onset (median age, 39 years vs 42 years; $P = .02$), truncal nodules (8% vs 3%; $P = .04$), rheumatoid arthritis (RA; 5% vs 1%; $P = .05$), and/or preceding tumor necrosis factor (TNF) inhibitor use (5% vs 1%; $P = .05$; **Tables 1** and **2**). Patients with chronic/recurrent EN compared with classic EN underwent more extensive evaluation for underlying causes, including autoantibodies (71% vs 55%; $P = .003$) and colonoscopy/endoscopy (14% vs 5%; $P = .003$). Underlying disease associations or triggers were identified in 256 patients (75%), with no difference between groups.

Patients with chronic/recurrent EN compared with classic EN more frequently required multiple therapies to achieve a clinical response (42% vs 25%; $P < .001$), and nonsteroidal immunomodulatory therapies, intralesional or topical corticosteroids, and potassium iodide were used more frequently to treat chronic/recurrent EN (Table 1). Systemic therapy was required in 64% of patients with chronic/recurrent EN vs 48% in patients with classic EN ($P = .003$), and incomplete therapeutic response occurred in 26% and 1%, respectively ($P < .001$).

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Table 2. Causes of Classic Erythema Nodosum (EN) and Chronic/Recurrent EN

Characteristic	No. (%) All patients	Classic EN	Chronic/ recurrent EN	*P* value
Total	340	181	159	NA
Idiopathic	84 (25)	45 (25)	39 (25)	.99
Autoimmune/inflammatory comorbidities				
Ankylosing spondylitis	2 (1)	1 (1)	1 (1)	.99
Behçet disease	8 (2)	2 (1)	6 (4)	.15
Chronic granulomatous mastitis	4 (1)	4 (2)	0	.06
Cutaneous lupus erythematosus	1 (0)	1 (1)	0	.99
Idiopathic inflammatory myopathy	1 (0)	1 (1)	0	.99
Inflammatory bowel disease	45 (13)	23 (13)	22 (14)	.76
Rheumatoid arthritis	10 (3)	2 (1)	8 (5)[a]	.05[b]
Sarcoidosis	30 (9)	15 (8)	15 (9)	.71
Systemic lupus erythematosus	9 (3)	4 (2)	5 (3)	.74
Systemic sclerosis	0	0	0	NA
Type 1 diabetes	1 (0)	0	1 (1)	.47
Undifferentiated connective tissue disease	6 (2)	4 (2)	2 (1)	.69
Any of the above	109 (32)	56 (31)	53 (33)	.64
Medication[c]				
Oral contraceptive or hormone replacement	49 (14)	25 (14)	24 (15)	.74
Systemic antimicrobial	52 (15)	33 (18)	19 (12)	.11
TNF inhibitor	10 (3)	2 (1)	8 (5)	.05
Vaccine	4 (1)	2 (1)	2 (1)	.99
New drug initiated <6 wk before symptom onset	15 (4)	6 (3)	9 (6)	.29
Infectious				
Endemic fungal infection	0	0	0	NA
Gastroenteritis	7 (2)	6 (3)	1 (1)	.13
History of hepatitis B	2 (1)	0	2 (1)	.22
History of hepatitis C	4 (1)	3 (2)	1 (1)	.63
History of HIV	0	0	0	NA
Respiratory tract infection, lower	4 (1)	3 (2)	1 (1)	.63
Respiratory tract infection, upper	66 (19)	39 (22)	27 (17)	.29
Skin and soft-tissue infection	9 (3)	4 (2)	5 (3)	.74
Tick-borne or Rickettsial disease	5 (1)	3 (2)	2 (1)	.99
Tuberculosis	9 (3)	2 (1)	7 (4)	.09
Urinary tract infection	10 (3)	5 (3)	5 (3)	.99
Vaginal infection	3 (1)	1 (1)	2 (1)	.60
Other infection	3 (1)	1 (1)	2 (1)	.60
Hematologic cancer	10 (3)	5 (3)	5 (3)	.99
Pregnancy	9 (3)	3 (2)	6 (4)	.31
Other (not classified above)[d]	25 (7)	15 (8)	10 (6)	.48

Abbreviations: NA, not applicable; TNF, tumor necrosis factor.

[a] Of these 8 patients with rheumatoid arthritis, only 2 had preceding TNF inhibitor use.

[b] There was a statistically significant association between rheumatoid arthritis and chronic/recurrent EN when adjusting for autoantibody testing ($P = .03$).

[c] Temporality was established between medication administration and symptom onset. Only medications given prior to symptom onset were included as potential triggers.

[d] Includes antiphospholipid syndrome, cholecystectomy, chronic autoimmune type nonspecific inflammatory arthritis, colitis, gout, head trauma, intestinal bacterial overgrowth, lung carcinoma, mastitis, pancreatic cancer, PASH (pyoderma gangrenosum, acne, and hidradenitis suppurativa) syndrome, pemphigus vulgaris, pericarditis, pernicious anemia, pituitary surgery, polymyalgia rheumatica, poorly differentiated adenocarcinoma, postpartum, ruptured breast implant, Schnitzler syndrome, Still disease, and undifferentiated seronegative spondyloarthropathy.

Discussion | To our knowledge, this cohort study is the largest to examine clinical characteristics of EN and the only US-based study examining chronicity/recurrence. This study highlights this topic's importance, as nearly half of all patients exhibited chronic/recurrent EN. Moreover, 26% of patients with chronic/recurrent EN demonstrated incomplete response to therapy despite EN's typical characterization as a self-resolving process.[4] Importantly, only 25% of cases in both groups were idiopathic, suggesting that evaluation for underlying triggers or comorbidities should be considered in all patients with EN, regardless of subtype. Further studies are needed to elucidate a potential association between EN, RA, and TNF inhibitors and to establish optimal treatment strategies for chronic/recurrent EN.

This retrospective, single-center study has limitations, including limited generalizability and that histopathology

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was not required for EN diagnosis, as EN is usually diagnosed clinically and requiring histopathology-confirmed disease may lead to sampling error. Potential misdiagnosis is also possible. Although this study has a large cohort, it may be insufficiently powered to identify subtle differences, such as racial differences, in those with chronic/recurrent EN. Further studies are warranted to identify risk factors leading to chronicity/recurrence of EN, results of autoantibody testing, and ideal management strategies.

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Accepted for Publication: November 1, 2023.

Published Online: January 3, 2024. doi:10.1001/jamadermatol.2023.5306

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Conflict of Interest Disclosures: Dr Femia reported royalties from UpToDate and consulting fees from Octagon Therapeutics, Timber Pharmaceuticals, and Guidepoint. No other disclosures were reported.

Funding/Support: Ms Shah is supported by the National Center for Advancing Translational Sciences of the National Institutes of Health through grant award number UL1TR001445.

Role of the Funder/Sponsor: The National Center for Advancing Translational Sciences had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Disclaimer: The content herein is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Data Sharing Statement: See the Supplement.

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Risk of Alopecia Areata After COVID-19

COVID-19 is associated with exacerbation or triggering of various autoimmune diseases, such as systemic lupus erythematosus, inflammatory bowel disease, and rheumatoid arthritis.[1] Alopecia areata (AA) is autoimmune hair loss that occurs in susceptible individuals by environmental triggers, such as viruses, vaccinations, and psychological stress. There is a growing number of reports on new onset, exacerbation, and recurrence of AA after COVID-19.[2-4] However, evidence supporting an association between COVID-19 and AA is limited.

Supplemental content

Methods | From October 8, 2020, to September 30, 2021, we conducted a propensity score–matched, nationwide, population-based cohort study to investigate the association between COVID-19 and AA using data from the Korea Disease Control and Prevention Agency–COVID-19–National Health Insurance Service cohort. Study population and methods are detailed in the eMethods in Supplement 1. This study was approved by the Jeonbuk National University Hospital (JBNUH) institutional review board, which waived informed consent due to use of deidentified data. The study followed the STROBE reporting guideline.

Incidence, prevalence, and adjusted hazard ratios (AHRs) for AA were calculated using the Cox proportional hazards regression model. The cohort was validated by examining positive and negative outcome controls. Significance was set at 2-sided $P < .05$. Analysis was performed using SAS, version 9.4 and R, version 4.0.3.

Results | The cohort comprised 259 369 patients with COVID-19 and 259 369 uninfected controls (**Table**). We found an increased risk of telogen effluvium in the cohort with COVID-19 compared with the uninfected cohort (AHR, 6.40; 95% CI, 4.92-8.33); incidence of negative control outcomes was not different between groups. Incidence of AA in patients with COVID-19 (43.19 per 10 000 person-years [PY]) was significantly higher than in controls (23.61 per 10 000 PY) regardless of clinical subtype (AHR, 1.82; 95% CI, 1.60-2.07) (**Figure**). Incidence of patchy AA and alopecia totalis and alopecia universalis (AT/AU) was 35.94 and 7.24 per 10 000 PY in patients with COVID-19 and 19.43 and 4.18 per 10 000 PY in controls, respectively. Incidence of AA after COVID-19 was significantly increased in all groups older than 20 years, with higher risk among both females and males. Prevalence of AA and AT/AU was 70.53 and 12.39 per 10 000 PY in patients with COVID-19, respectively—significantly higher than in controls (52.37 and 8.97 per 10 000 PY, respectively). The association between COVID-19 and AA was supported by sensitivity analysis with matching variables.

Discussion | A Korean population-based study did not find an association between COVID-19 and AA.[5] However, that study was small and limited to early stages of the pandemic. In this study, we found a significant increase in the incidence and prevalence of AA after COVID-19 even after adjusting for several confounding factors. During the study period, the age- and sex-adjusted in-

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